Redis Assignment

In this assignment, you will access a redis server and user redis commands to find out answers. The redis server is at *lab.aimet.tech*. You also have to authenticate as username 'hw' with password 'hw'.

The populated data in the redis database is similar to the example "simple social network" in the class. Answer all questions in mycourseville assignment.

Note that this user can only use "read" commands e.g. "get", "lrange", "llen", "scan", etc.

```
# we will have to install redis in colab
import sys
IN_COLAB = 'google.colab' in sys.modules
if IN_COLAB:
  !pip install redis
```

```
Collecting redis
  Downloading redis-5.0.1-py3-none-any.whl (250 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 250.3/250.3 kB 2.7 MB/s eta 0:00:00
Requirement already satisfied: async-timeout>=4.0.2 in /usr/local/lib/python3.10/dist-packages (from redis) (4.0.3)
Installing collected packages: redis
Successfully installed redis-5.0.1
```

```
import redis
```

```
rd = redis.Redis(host='lab.aimet.tech', charset="utf-8", decode_responses=True)
rd.auth(username='hw', password='hw')

# show all keys
print(rd.keys('*'))
```

```
['user:547:follows', 'username:shamefulOil5', 'user:666:follows', 'user:519:name', 'user:682:follows', 'username:humorousChough1', 'use
```

What is the username of user id "600"?

```
rd.get('user:600:name')
```

```
'cautiousCrackers9'
```

What is the id of username "excitedPie4" ?

```
rd.get('username:excitedPie4')
```

```
'567'
```

How many users that "excitedPie4" follows ?

```
rd.scard('user:567:follows')
```

```
9
```

How many users are there in the database?

```
numberOfUsers = sum([1 for _ in rd.keys('username:*')])

print("Number of users: ", numberOfUsers)
```

```
Number of users:  200
```

What is the average number of follows per user?

```
followers = [rd.scard(key) for key in rd.keys('user:*:follows')]

print("Average number of followers: ", sum(followers) / numberOfUsers)
```

```
Average number of followers:  8.605
```

How many users follows between 5-10 users?

```
users = [1 for key in rd.keys('user:*:follows') if rd.scard(key) >= 5 and rd.scard(key) <= 10]

print("Number of users follows between 5-10 users: " ,sum(users))
```

```
Number of users follows between 5-10 users:  60
```

Which account has the most followers?

```
maxUser = max(rd.keys('user:*:followed_by'), key=lambda x: rd.scard(x))

print("User ID: ", maxUser.split(":")[1])
```

```
User ID:  630
```